prop:N2-to-G2 in proof-theory/normalization

The Open Logic Project contributors

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Editorial note. All mathematical statement and proof text is reproduced verbatim from the Open Logic Project, unit proof-theory, file translations.tex, fixed at commit 1e960beff9ed7835bf3e3f1335e21af3439cd107. Only the packaging (the adapter preamble and the standalone wrapper) is this module's.

Editorial note. Numbering inside this standalone excerpt is editorial, not the Open Logic Project print numbering.

Editorial note. The adapter preamble loads the project's OWN style layer verbatim (sty/open-logic-formulas.sty, sty/open-logic-tokenize.sty, sty/open-logic-selective.sty and open-logic-config.sty, including its tag truth assignment), so the author's text-token shorthand, !A math metavariables, rule names (Intro, Elim, FalseInt) and the tag conditionals resolve exactly as in the book. Three substitutions are forced by the pinned toolchain and are recorded: isomath.sty is absent so \mathsfit is \mathsf; xpunctuate.sty is absent and unused so its RequirePackage is dropped; thmtools/ntheorem are absent so the environments are amsthm with the project's family numbering. The proof trees are the project's own bussproofs-extra.

Editorial note. The H2 judgement and its reason were made in this module by reading the author's own statement and complete proof at the pinned line span. Original candidate key: proof-theory/normalization|translations.tex|prop:normal-N2i-to-G2i.

Editorial note. Reference labels in the proof resolve to navigation pointers to the original author source; these pointers do not reproduce proofs of the prerequisites. The author's own ollabel is kept unprefixed (not part:chapter:section) so those pointers resolve; the label text itself is the author's own argument.

Allowed standard prerequisites (editorial).

- Prerequisites named by the author inside the counted span. The counted span is reproduced verbatim, including the author's own olref pointers. Its prerequisites are the earlier results of the same pinned Open Logic Project commit that the author cites there.

Source reference. Original reference 0000: sources/translation-N2i.tex, source lines 13–13. Navigation pointer used in the counted span; the span contains the pointer only and the cited result is not reproduced inside it. Registered as a non-load-bearing navigation stub under the name the author's own pointer produces. No bibliography citation is involved.

Proposition 0.1. *If δ is a normal* **N2c** *or* **N2i***-derivation of $\Gamma \Rightarrow \varphi$ then there is a (cut-free)* **G2c***-derivation (***G2i***-derivation) π of $\Gamma' \Rightarrow \Delta$, where Γ' is the multiset of formulas resulting from Γ by removing labels, and $\Delta = \{\varphi\}$ if φ is not $\bot$, and $\Delta = \emptyset$ if it is.*

Proof. This time, induction is on the number of inferences in the derivation δ of $\Gamma \Rightarrow \varphi$. If there is no inference in δ, it is an axiom $x : \varphi \Rightarrow \varphi$, and π is the corresponding axiom $\varphi \Rightarrow \varphi$ or $\bot \Rightarrow$ if $\varphi \equiv \bot$.

If δ ends in an inference rule, we distinguish cases:

1. If R is an introduction rule or $\bot_I$, the translations proceeds exactly as in the proof of 0000, and does not require Cut.

   If the last rule R of δ is an elimination rule, observe the following:

   (a) A main branch of δ passes through no introduction rules, as otherwise there would be an introduction rule or $\bot_I$ followed by an elimination rule along the main branch, and δ would not be normal.

   (b) Since there are no introduction rules along main branches of δ, there is only one main branch. (Only $\wedge$Intro has two major premises, so multiple main branches would have to go through the premises of a $\wedge$Intro rule.)

   (c) If the main branch includes the major premise of $\vee$Elim or $\exists$Elim, there is only one such rule and it is the last one, i.e., R. (Otherwise the main branch would contain an $\vee$Elim or $\exists$Elim rule where the conclusion of the rule is major premise of an elimination rule, and the proof would not be normal as we could apply a permutation conversion.)

   (d) No elimination rules other than $\vee$Elim and $\exists$Elim discharge assumptions, i.e., change the left context of sequents. $\vee$Elim and $\exists$Elim only discharge assumptions above a minor premise, i.e., change the left context only of their minor premises. In other words, if a sequent S_i on the main branch of δ has context Γ_1, the conclusion S_{i+1} of the inference of which S_i is the major premise has context $\Gamma_1' \supseteq \Gamma_1$.

   (e) Consequently, if $x : \chi \Rightarrow \chi$ is the topmost sequent S_1 in the main branch, $x : \chi \in \Gamma$.

   We now distiguish cases according to the form of χ. Since every S_i in the main branch is major premise of an elimination rule, this also applies to $x : \chi \Rightarrow \chi$.

2. $\chi \equiv \theta \wedge \alpha$. Then δ has the form

$$\dfrac{\dfrac{x : \theta \wedge \alpha \Rightarrow \theta \wedge \alpha}{x : \theta \wedge \alpha \Rightarrow \theta}\,\wedge\mathrm{Elim}_1}{\begin{array}{c}\vdots\\ x : \theta \wedge \alpha, \Gamma_1 \Rightarrow \varphi\end{array}}$$

   The main branch includes $x : \theta \wedge \alpha \Rightarrow \theta$ contains no major premise of $\rightarrow$Intro, $\neg$Intro nor a minor premise of $\vee$Elim or $\exists$Elim, the context

formulas in $x : \theta \wedge \alpha$ remain unchanged along the main branch. This explains why the context of the end-sequent must contain $x : \theta \wedge \alpha$. Moreover, we can turn the derivation δ into a derivation δ_1 by replacing the sub-derivation ending in $\wedge$Elim by $x : \theta \Rightarrow \theta$ and replacing $x : \theta \wedge \alpha$ in the context of every sequent along the main branch by $x : \theta$, i.e., turn

$$\dfrac{\dfrac{x : \theta \wedge \alpha \;\Rightarrow\; \theta \wedge \alpha}{x : \theta \wedge \alpha \;\Rightarrow\; \theta}\,\wedge\mathrm{Elim}_1}{\vdots}\qquad\qquad \vdots$$
$$x : \theta \wedge \alpha, \Gamma_1 \;\Rightarrow\; \varphi \qquad \text{into} \qquad x : \theta, \Gamma_1 \;\Rightarrow\; \varphi$$

The induction hypothesis applies to δ_1, since the number of inferences in δ is the number of inferences in δ' plus 1.

By induction hypothesis, there is a **G2i**-derivation π_1 of $x : \theta, \Gamma_1 \Rightarrow \varphi$. We obtain the derivation π by applying $\wedge$L:

$$\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \theta, \Gamma_1' \;\Rightarrow\; \varphi\end{array}}{\theta \wedge \alpha, \Gamma_1' \;\Rightarrow\; \varphi}\,\wedge\mathrm{L}$$

If $\varphi \equiv \bot$ we insert a WR, e.g,

$$\dfrac{\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \theta, \Gamma_1' \;\Rightarrow\end{array}}{\theta \wedge \alpha, \Gamma_1' \;\Rightarrow}\,\wedge\mathrm{L}}{\theta \wedge \alpha, \Gamma_1' \;\Rightarrow\; \varphi}\,\mathrm{WR}$$

3. If $\chi \equiv \theta \vee \alpha$, it must be the major premise of $\vee$Elim, and this inference must be the last inference R on the main branch. In other words, δ has the form:

$$\dfrac{x : \theta \vee \alpha \;\Rightarrow\; \theta \vee \alpha \qquad \begin{array}{c}\vdots\,\delta_1\\ y : \theta, \Gamma_1 \;\Rightarrow\; \varphi\end{array} \qquad \begin{array}{c}\vdots\,\delta_2\\ z : \alpha, \Gamma_2 \;\Rightarrow\; \varphi\end{array}}{x : \theta \vee \alpha, \Gamma_1, \Gamma_2 \;\Rightarrow\; \varphi}\,\vee\mathrm{Elim}$$

The induction hypothesis applies to the derivations δ_1 and δ_2; we obtain **G2i**-derivations π_1 and π_2 of $\theta, \Gamma_1' \Rightarrow \varphi$ and $\alpha, \Gamma_2' \Rightarrow \varphi$. We apply $\vee$L to get π:

$$\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \theta, \Gamma_1' \;\Rightarrow\; \varphi\end{array} \qquad \begin{array}{c}\vdots\,\pi_2\\ \alpha, \Gamma_2' \;\Rightarrow\; \varphi\end{array}}{\theta \vee \alpha, \Gamma_1', \Gamma_2' \;\Rightarrow\; \varphi}\,\vee\mathrm{L}$$

Should $\vee$Elim not discharge $y : \theta$ or $z : \alpha$ (i.e., these aren't present in the context of the minor premises), or should φ happen to be $\bot$, we have to

add some WL and WR. For instance,

$$\dfrac{\dfrac{\dfrac{\vdots\,\pi_1}{\Gamma_1' \Rightarrow}}{\theta, \Gamma_1' \Rightarrow}\,\mathrm{WL} \qquad \dfrac{\dfrac{\vdots\,\pi_2}{\Gamma_2' \Rightarrow}}{\alpha, \Gamma_2' \Rightarrow}\,\mathrm{WL}}{\theta \vee \alpha, \Gamma_1', \Gamma_2' \Rightarrow}\,\vee\mathrm{L}}{\theta \vee \alpha, \Gamma_1', \Gamma_2' \Rightarrow \varphi}\,\mathrm{WR}$$

4. $\chi \equiv \theta \to \alpha$. Then δ has the form

$$\dfrac{\dfrac{x : \theta \to \alpha \Rightarrow \theta \to \alpha \qquad \dfrac{\vdots\,\delta_1}{\Gamma_1 \Rightarrow \theta}}{x : \theta \to \alpha, \Gamma_1 \Rightarrow \alpha}\,{\to}\mathrm{Elim} \quad \vdots\,\delta_2}{x : \theta \to \alpha, \Gamma_1, \Gamma_2 \Rightarrow \varphi}$$

Here note that since the main branch, which includes $x : \theta \to \alpha, \Gamma_1 \Rightarrow \alpha$, contains no major premise of $\to$Intro, $\neg$Intro nor a minor premise of $\vee$Elim or $\exists$Elim, the context formulas in $x : \theta \to \alpha, \Gamma_1$ remain unchanged along the main branch. This explains why the context of the end-sequent must contain $x : \theta \to \alpha, \Gamma_1$. Moreover, we can turn the derivation fragment δ_2 into a correct derivation δ_2' by replacing the sub-derivation ending in $\to$Elim by $x : \alpha \Rightarrow \alpha$ and replacing $x : \theta \to \alpha, \Gamma_1$ in the context of every sequent along the main branch by $x : \alpha$, i.e., turn

$$\dfrac{x : \theta \to \alpha, \Gamma_1 \Rightarrow \alpha \quad \vdots\,\delta_2}{x : \theta \to \alpha, \Gamma_1, \Gamma_2 \Rightarrow \varphi} \quad \text{into} \quad \dfrac{x : \alpha \Rightarrow \alpha \quad \vdots\,\delta_2'}{x : \alpha, \Gamma_2 \Rightarrow \varphi}$$

The induction hypothesis applies to δ_1 and δ_2', since the number of inferences in δ is the sum of the numbers of inferences in δ_1 and δ_2, plus 1 to account for the $\to$Elim inference at the start of the main branch. (If that inference is also the last inference R in δ then δ_1 contains 0 inferences.)

By induction hypothesis, there are **G2i**-derivations π_1 of $\Gamma_1 \Rightarrow \theta$ and π_2 of $\alpha, \Gamma_1' \Rightarrow \varphi$. We obtain the derivation π by applying $\to$L:

$$\dfrac{\dfrac{\vdots\,\pi_1}{\Gamma_1' \Rightarrow \theta} \qquad \dfrac{\vdots\,\pi_2}{\alpha, \Gamma_2' \Rightarrow \varphi}}{\theta \to \alpha, \Gamma_1', \Gamma_2' \Rightarrow \varphi}\,{\to}\mathrm{L}$$

If θ and/or $\varphi \equiv \bot$ we insert one or two WR, e.g,

$$\dfrac{\dfrac{\begin{array}{c}\vdots\,\pi_1\\ \Gamma_1' \Rightarrow\end{array}}{\Gamma_1' \Rightarrow \theta}\,\mathrm{WR} \qquad \dfrac{\begin{array}{c}\vdots\,\pi_2\\ \alpha, \Gamma_2' \Rightarrow\end{array}}{\alpha, \Gamma_2' \Rightarrow \varphi}\,\mathrm{WR}}{\theta \to \alpha, \Gamma_1', \Gamma_2' \Rightarrow \varphi}\,{\to}\mathrm{L}$$

The remaining cases are treated similarly and are left as exercises. □